

Strategies for high quality standard spectra

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Standards

- Standard spectra are spectra collected from materials of known composition under well controlled measurement conditions.
- Standard spectra provide the x-ray intensity data against which spectra from unknown materials will be compared.
- You will want a standard for each element in the unknown*.
- One spectrum can potentially serve as a standard for more than one element.

* With the exception of elements which may be deduced from stoichiometry or another assumption.

The quality of standard spectra is critical!

A poor quality standard will cause damage every time it is used. It will damage the results for that element and all the other measured elements.

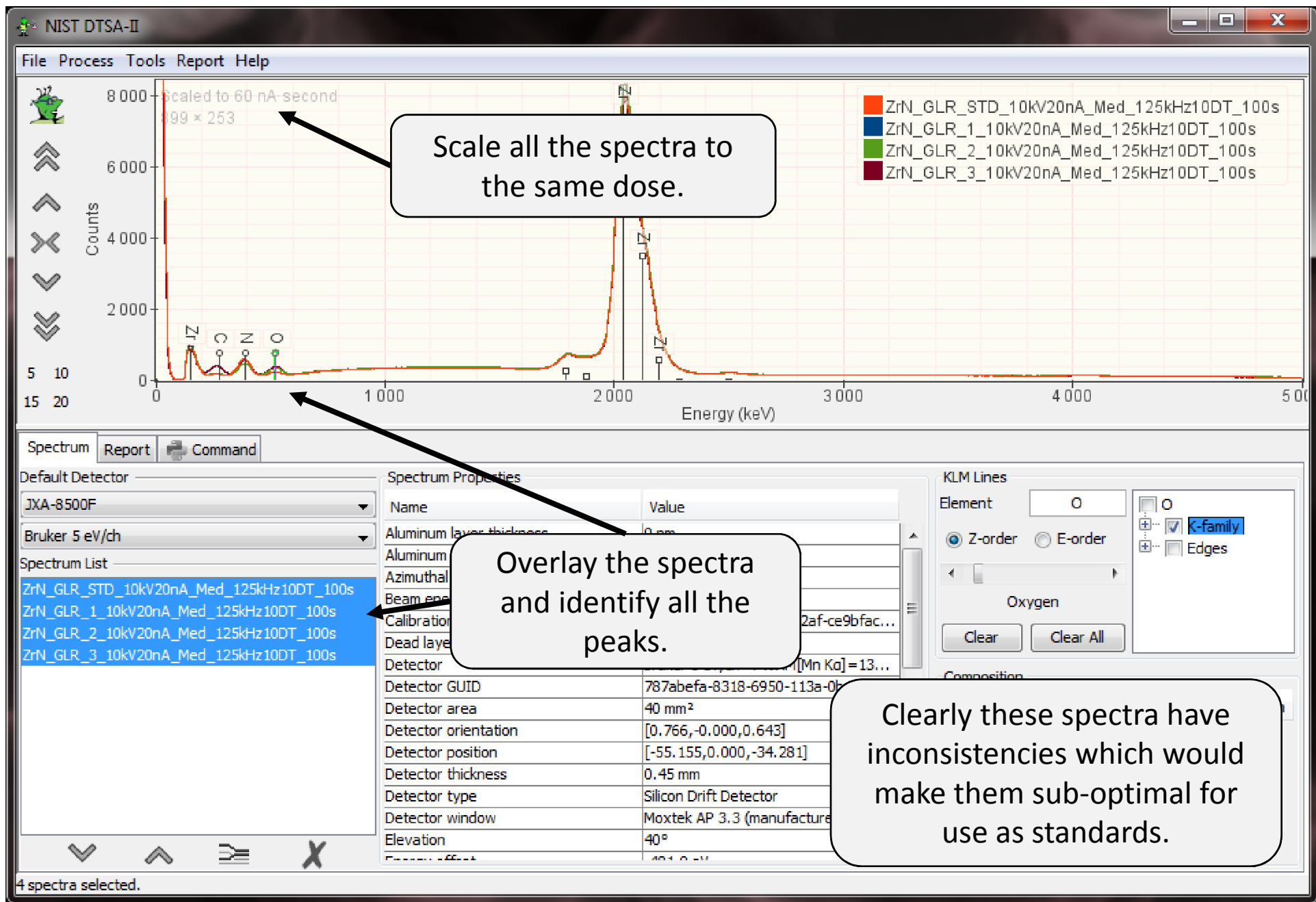
Strategy

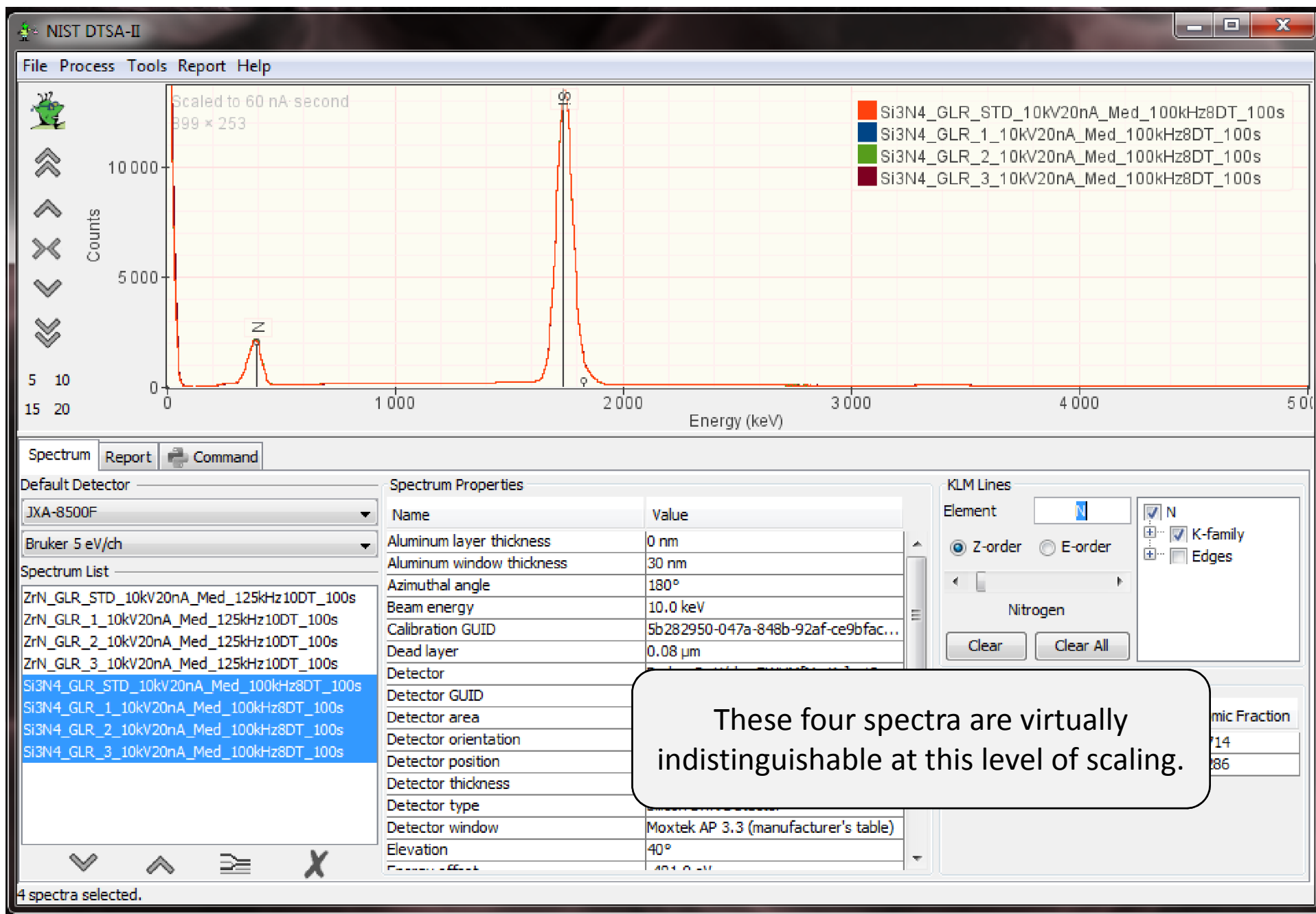
- Collect multiple (4 or more) replica standard spectra and compare them one to another to determine if any are outliers.
 - Look for changes in intensity in all the peaks associated with the element.
 - Changes may suggest topographic effects
 - Changes may suggest inhomogeneity
 - Look for additional peaks
 - Additional peaks may suggest contamination
 - Additional peaks may suggest inhomogeneity
 - Look for changes in the Bremsstrahlung / continuum background.

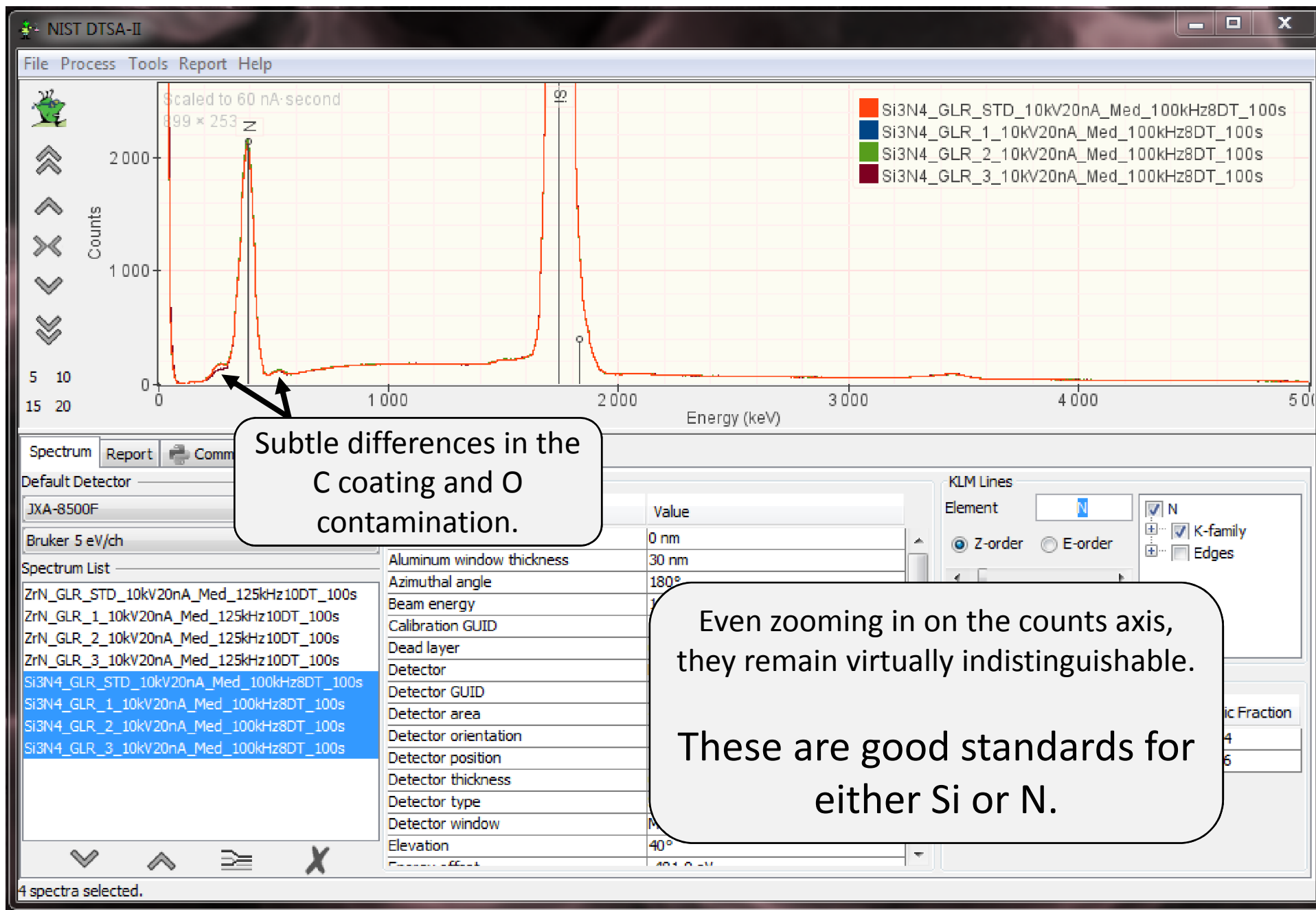
Any variation suggests a problem.

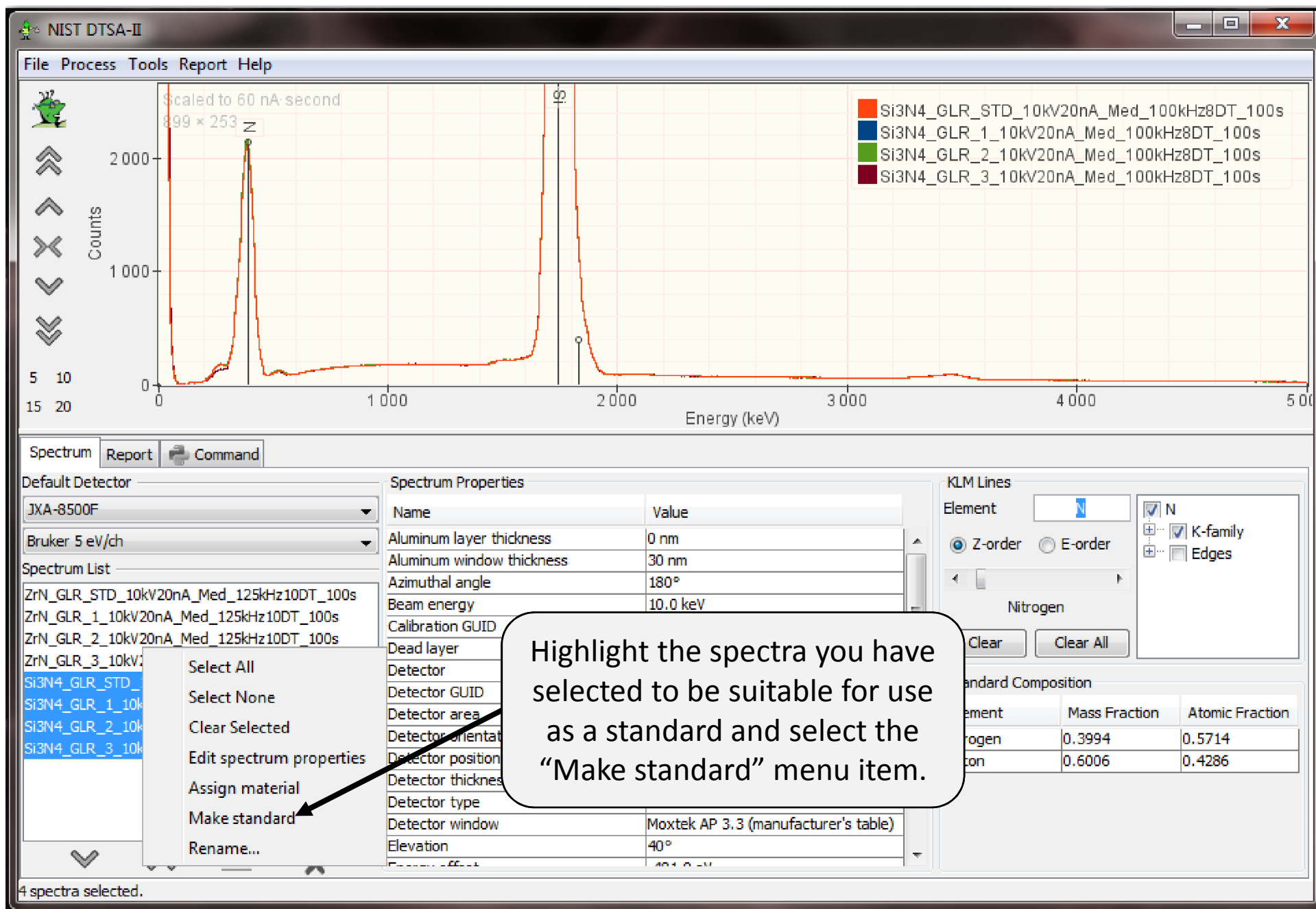
Strategy

- Discard outlier spectra
 - Insist that at least 3 of the 4 spectra are indistinguishable except for routine variation due to count statistics.
- Sum together the consistent spectra to form a single high quality spectrum.
 - Improves count statistics by summing the dose.
 - Averages out any subtle differences.









The "Make standard" dialog automatically makes many sanity checks to ensure that the spectra are compatible and all the necessary information is provided.

The beam energy is extracted from the spectra.

Specify the material from which the spectra were collected.

The detector associated with all the spectra. All the spectra must be collected on the same detector.

The probe dose is the sum of the doses from the individual spectra. If a spectrum is missing either the live time or the probe current you will be asked to provide it.

The "Save" button saves the resulting spectrum directly to an EMSA file.

Selecting "Ok" adds the new standard to the Spectrum List.

4 spectra selected.

Legend:

- Si3N4_GLR_STD_10kV20nA_Med_100kHz8DT_100s
- Si3N4_GLR_1_10kV20nA_Med_100kHz8DT_100s
- Si3N4_GLR_2_10kV20nA_Med_100kHz8DT_100s
- Si3N4_GLR_3_10kV20nA_Med_100kHz8DT_100s

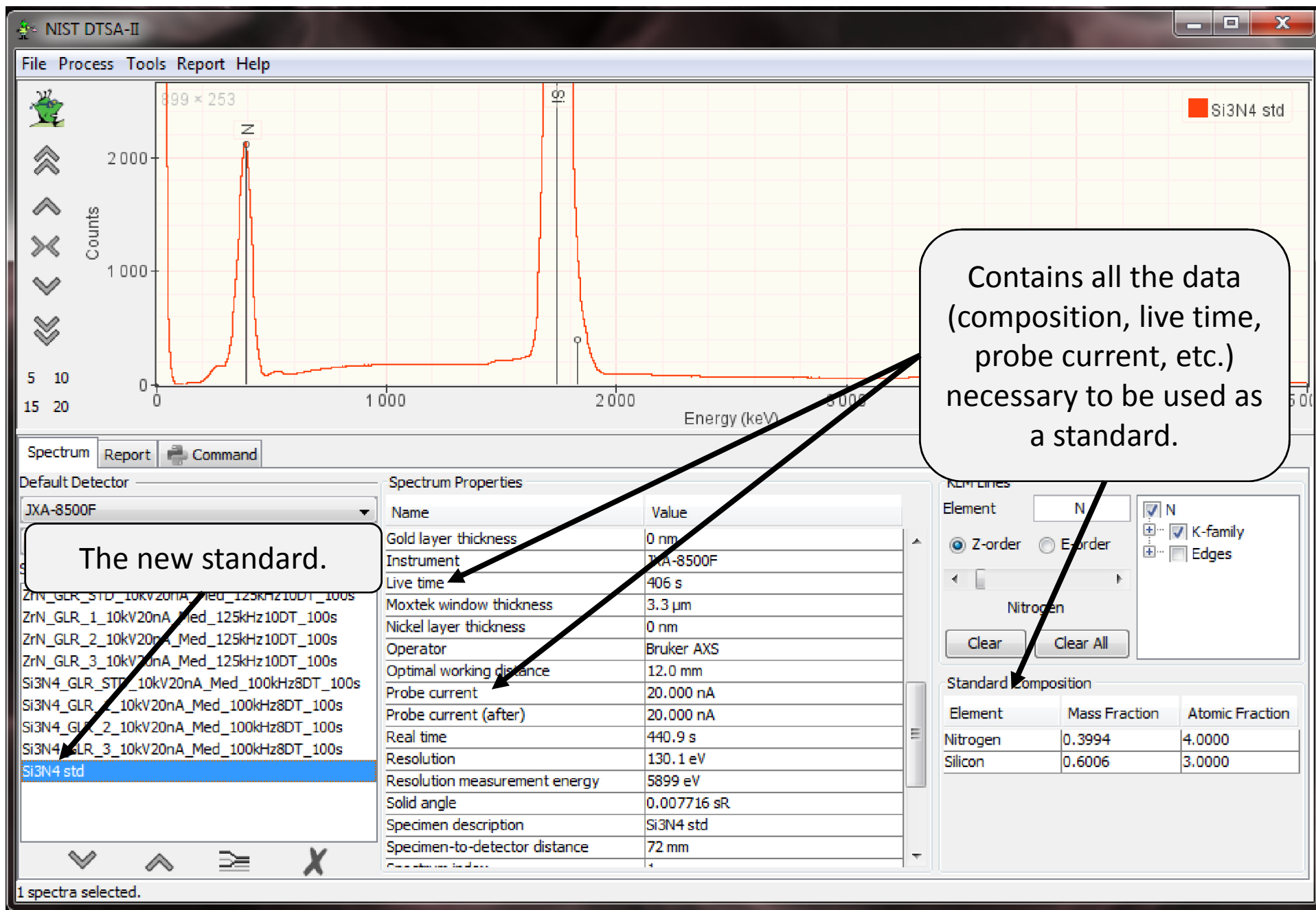
Make standard dialog fields:

- Material: Si3N4
- Beam energy: 10.0 keV
- Probe Dose: 8119.4 nA*s
- Detector: Bruker 5 eV/ch

Buttons: Save, Ok, Cancel

Background interface elements:

- File, Pr...
- Spectrum, Report, Command
- Default Detector: JXA-8500F
- Spec...
- Table with columns: Name, Minimum, Maximum, muthal, am ene, libration GUID, ad layer, detector, detector GUID
- Standard Composition



Dirty Trick? 1

Sometimes a spectrum will almost be a perfect standard except for the presence of an ill-placed peak. Use the Trim tool to eliminate the offending peak.

The screenshot shows the NIST DTSA-II software interface. The main window displays a spectrum plot with a red line representing the data. A yellow vertical bar highlights a specific region of the spectrum, and a black arrow points from the 'Trim' option in the 'Process' menu to this region. The 'Process' menu is open, showing options like 'Sub-sample spectrum', 'Fit background', 'Strip background', 'Linearize energy axis', 'Smooth (Savitzky-Golay)', 'Trim', and 'Peak search'. The 'Trim' option is highlighted. The 'Spectrum List' on the left shows a list of spectra, with 'Si3N4 std' selected. The 'Spectrum Properties' table on the right provides details about the selected spectrum. The 'KLM Lines' section shows the element 'N' (Nitrogen) and the 'Standard Composition' table.

File Process Tools Report Help

Sub-sample spectrum
Fit background
Strip background
Linearize energy axis
Smooth (Savitzky-Golay)
Trim
Peak search

200
0
0 1000

5 10
15 20

Spectrum Report Command

Default Detector
JXA-8500F
Bruker 5 eV/ch

Spectrum List

- ZrN_GLR_STD_10kV20nA_Med_125kHz10DT_100s
- ZrN_GLR_1_10kV20nA_Med_125kHz10DT_100s
- ZrN_GLR_2_10kV20nA_Med_125kHz10DT_100s
- ZrN_GLR_3_10kV20nA_Med_125kHz10DT_100s
- Si3N4_GLR_STD_10kV20nA_Med_100kHz8DT_100s
- Si3N4_GLR_1_10kV20nA_Med_100kHz8DT_100s
- Si3N4_GLR_2_10kV20nA_Med_100kHz8DT_100s
- Si3N4_GLR_3_10kV20nA_Med_100kHz8DT_100s
- Si3N4 std

Spectrum Properties

Name	Value
Gold layer thickness	0 nm
Instrument	JXA-8500F
Live time	406 s
Moxtek window thickness	3.3 μ m
Nickel layer thickness	0 nm
Operator	Bruker AXS
Optimal working distance	12.0 mm
Probe current	20.000 nA
Probe current (after)	20.000 nA
Real time	440.9 s
Resolution	130.1 eV
Resolution measurement energy	5899 eV
Solid angle	0.007716 sr
Specimen description	Si3N4 std
Specimen-to-detector distance	72 mm

KLM Lines

Element N

☒ N
☒ K-family
☐ Edges

☒ Z-order ☐ E-order

Nitrogen

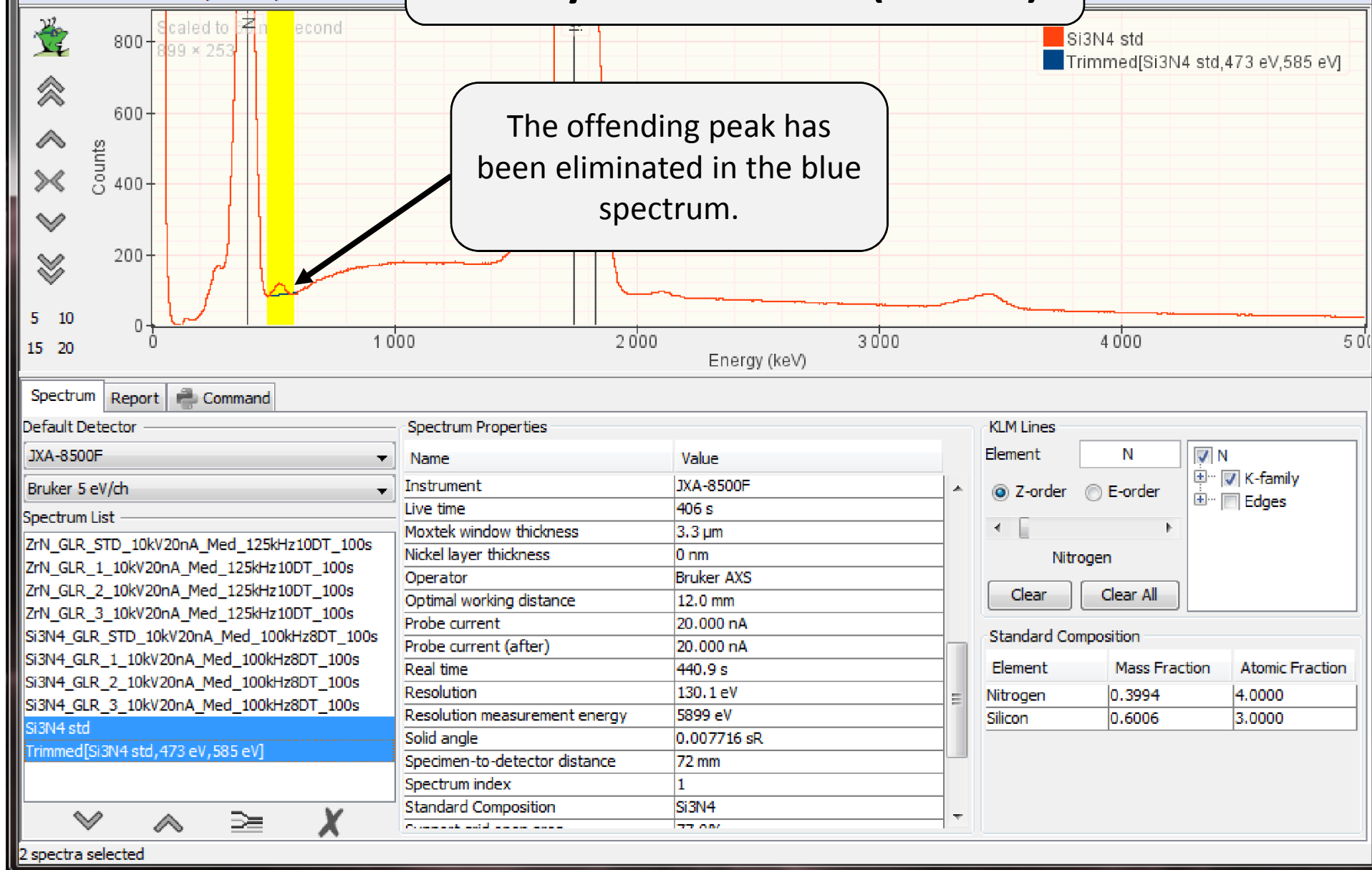
Clear Clear All

Standard Composition

Element	Mass Fraction	Atomic Fraction
Nitrogen	0.3994	4.0000
Silicon	0.6006	3.0000

1 spectra selected.

Dirty Trick? 1 (cont)



Dirty Trick? 2

NIST DTSA-II

File Process Tools Report Help

Counts

899 × 253

Si3N4 std

Calibrate an EDS detector

Measured spectrum

Specify a spectrum

Spectrum Si3N4 std

Specify the material

Material Si3N4OC

Live time 405.97 sec. Probe current 20.000 nA

Fit type Linear (default)

Specify an effective date

Effective date Dec 31, 1969

More...

Back Next Finish Cancel

Trimmed[Si3N4 std, 473 eV, 585 eV]

Resolution measurement energy 5899 eV

Solid angle 0.007716 sr

Specimen description Si3N4 std

Specimen-to-detector distance 72 mm

1 spectra selected.

Use the calibration alien to fit all the elements in the spectrum.

Add small amounts of C and O to the composition to force the algorithm to fit these elements.

Mass Fraction Atomic Fraction

Mass Fraction	Atomic Fraction
0.3994	4.0000
0.6006	3.0000

Dirty Trick? 2 (cont)

NIST DTSA-II

File Process Tools Report Help

Scaled to 899 x 253

Counts

800

600

400

200

0

0

5 10 15 20

Spectrum Report Command

Default Detector

JXA-8500F

Bruker 5 eV/ch

Spectrum List

- ZrN_GLR_STD_10kV20nA
- ZrN_GLR_1_10kV20nA
- ZrN_GLR_2_10kV20nA
- ZrN_GLR_3_10kV20nA_Med_100kHz8DT
- Si3N4_GLR_STD_10kV20nA_Med_100kHz8DT
- Si3N4_GLR_1_10kV20nA_Med_100kHz8DT
- Si3N4_GLR_2_10kV20nA_Med_100kHz8DT
- Si3N4_GLR_3_10kV20nA_Med_100kHz8DT
- Si3N4 std
- Trimmed[Si3N4 std,473 eV,585 eV]
- Si3N4 std

5 spectra selected

Calibrate an EDS detector

Previous: Fit Results

Fit results

Finish

Energy calibration

Fano factor 0.1200

Noise 5.95±0.00 eV

FWHM 129.96±0.01 eV

Resolution calibration

Zero offset -459.19±0.03 eV

Channel width 4.9597±0.0001 eV/channel

☐ Add to database

☒ Output elemental fits

Message:

More...

Back Next Finish Cancel

Select "Output elemental fits."

Deselect "Add to database"

Si3N4 std

Si3N4 std

Fit[Si3N4 std]

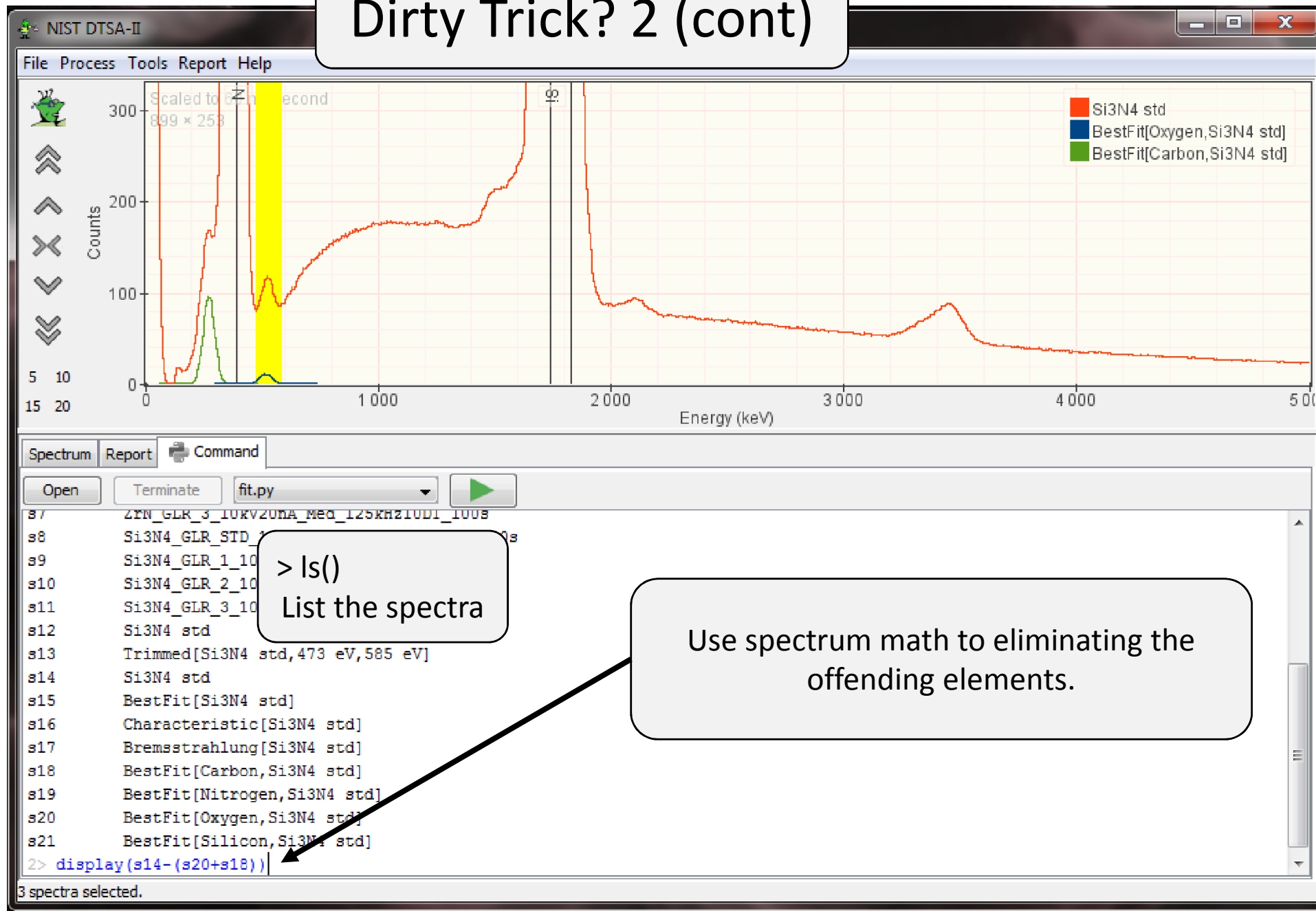
Characteristic[Si3N4 std]

Scattering[Si3N4 std]

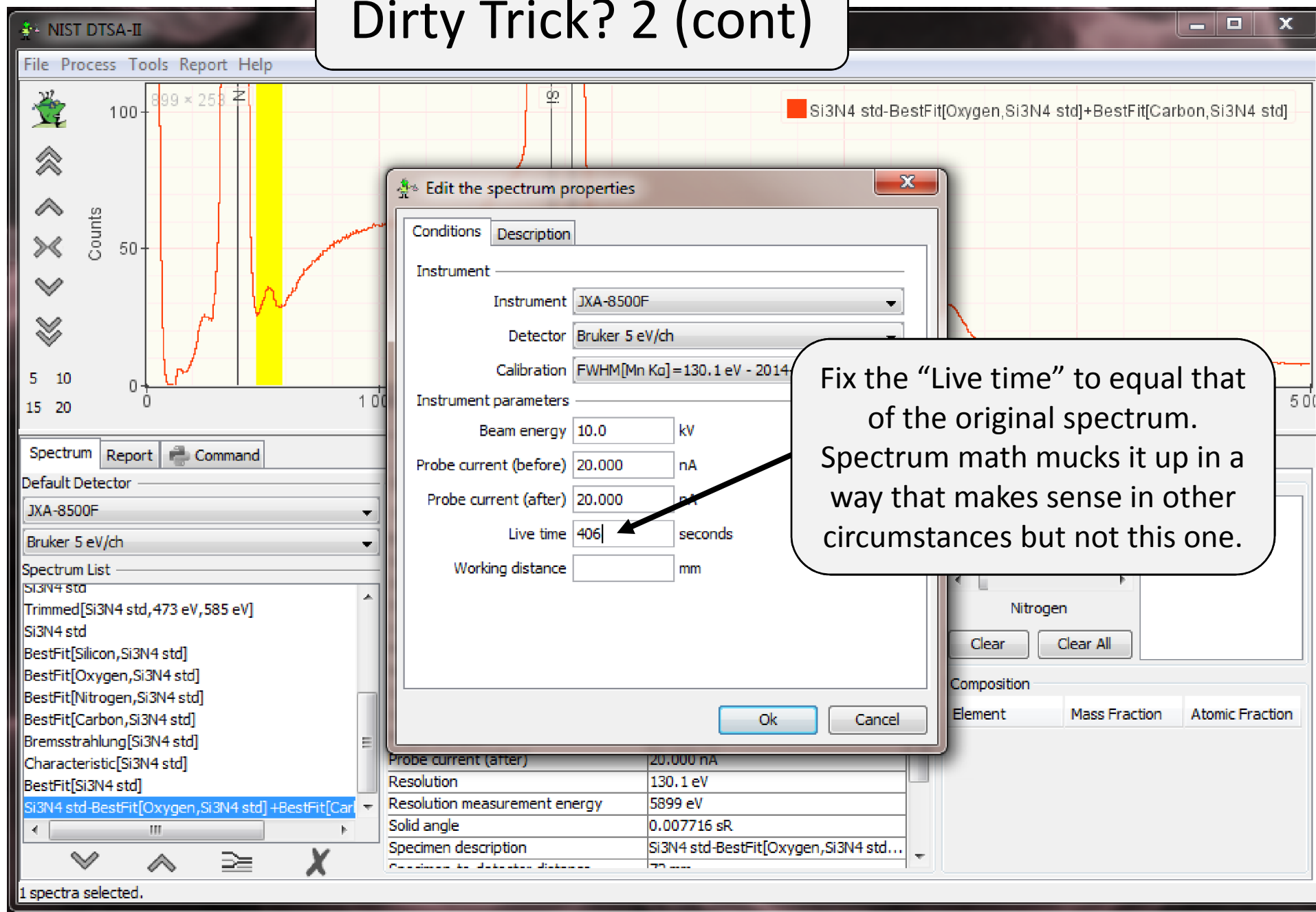
fraction Atomic Fraction

Spectrum index	1
Support grid open area	77.0%
Support grid thickness	0.380 mm
Window type	Ultra-thin window

Dirty Trick? 2 (cont)



Dirty Trick? 2 (cont)



Dirty Trick? 2A

